# EXP-006 — Contextual Enrichment Ablation

Isolating structural context from chunk size, to find out which half of the previous experiment's gain was real.

EXECUTIVE RESULT

**Contextual enrichment did not improve retrieval.** On the control chunking, adding a structural header to the indexed text moved macro span recall by **exactly zero** (0.475 → 0.475) — one question rescued, one regressed.

The measured reason: enrichment **inflates document frequency**. Writing `Provider: anthropic` into all 12,028 Anthropic chunks took the term "Anthropic" from df 3,289 to 12,028, destroying its IDF. Across 22 evidence spans, enrichment supplied a query term to only 3, and in none of those was it a discriminative one.

**Δ 0.000**

Macro recall change from enrichment on control chunking (A→B) — the comparison that isolates enrichment

**3 of 22**

Evidence spans where enrichment supplied any query term at all; zero of them discriminative

## The 2×2

B and D are row-for-row copies of A and C — **0** boundary differences and **0** body differences, verified in SQL and pinned by tests. Only the indexed text differs, so an A→B difference cannot be a chunking difference. The canonical chunk body is never mutated: the header lives in a separate `search_text` column, because a citation must quote real source text.

| Configuration | macro recall | fully recalled | spans @10 | doc recall | MRR | absent@10 | @50 | @100 | @300 | ms |
|---|---|---|---|---|---|---|---|---|---|---|
| A — control chunking, plain | 0.475 | 9/20 | 10/22 | 0.825 | 0.280 | 12 | 3 | 2 | 1 | 418 |
| B — control chunking, enriched | 0.475 | 9/20 | 10/22 | 0.875 | 0.282 | 12 | 3 | 2 | 1 | 513 |
| C — bounded chunking, plain | 0.500 | 9/20 | 11/22 | 0.825 | 0.287 | 11 | 5 | 4 | 1 | 446 |
| **D — bounded chunking, enriched** | 0.550 | 10/20 | 12/22 | 0.850 | 0.279 | 10 | 3 | 2 | 0 | 586 |

## Paired comparisons

| Comparison | Δ macro | rescued | regressed | net |
|---|---|---|---|---|
| A->B enrichment on control chunking | +0.000 | AN-004 | AN-005 | +0 |

| Comparison | Δ macro | rescued | regressed | net |
|---|---|---|---|---|
| A->C chunk size without enrichment | +0.025 | — | — | +0 |
| C->D enrichment on bounded chunking | +0.050 | AN-004, AN-007 | AN-005 | +1 |
| B->D chunk size with enrichment | +0.075 | AN-007 | — | +1 |

**A→B is the headline and it is a wash.** Its single rescue (AN-004, rank 12→7) crossed the cutoff while its BM25 score went *down* — competitors simply lost more. Its regression (AN-005, rank 4→18) is large and mechanistic: the query's best term `editing` had df 66, and writing the section heading into every chunk of that document tripled it to 211.

## Why enrichment nets to zero

A constant field repeated across every chunk of a document or provider is, by construction, non-discriminative. BM25 weights a term by `ln(1 + (N-df+0.5)/(df+0.5))`, so tripling a term's df directly attacks the signal that made it useful.

| term | df plain | df enriched | change |
|---|---|---|---|
| Anthropic | 3,289 | 12,028 | **+266%** |
| OpenAI | 366 | 2,185 | **+497%** |
| beta | 1,405 | 2,933 | **+109%** |
| editing | 66 | 211 | **+220%** |

98 distinct query terms shifted document frequency. Enrichment adds a handful of weak matches while diluting terms that were already working.

## AN-003 — the canonical failure

| Configuration | rank | in top 300? | chunk chars | doc rank |
|---|---|---|---|---|
| A — control chunking, plain | — | no | — | 6 |
| B — control chunking, enriched | — | no | — | 9 |
| C — bounded chunking, plain | — | no | — | 4 |
| D — bounded chunking, enriched | 74 | yes | 1191 | 12 |

Enrichment built a lexical bridge for the first time — AN-003's evidence becomes reachable at **rank 74** under D, having been absent at depth 300 in every previous experiment. The header supplied `Batches`, a term the body lacked. But enrichment simultaneously took that term's df from 283 to 1,203, and the terms that would actually identify this evidence remain absent everywhere: `contain` (df 111), `most` (533), `many` (564), and `requests` — which never matches the body's singular `request`.

**Answer: no.** Section-path enrichment does not create enough of a lexical bridge to make AN-003 retrievable. Rank 74 is still a failure at k=10 and would remain one at k=50.

## Was enrichment the real source of V3's gain?

**No.** V3 rescued AN-004, AN-008 and AN-010 in EXP-005. Enrichment alone rescues AN-004 only — fragilely — while losing AN-005. Enrichment on bounded chunking rescues AN-004 and AN-007, but *neither* AN-008 nor AN-010. AN-010 was V3's largest win (rank 49→1) and no EXP-006 configuration reproduces it, so that rescue must have come from V3's table row-group splitting, which EXP-006 did not test.

## Exploratory: which header fields matter

Run after the core 2×2, motivated by the df-inflation mechanism. **Selected on the development set — not a held-out result.**

| Variant | Header fields | macro recall | fully recalled |
|---|---|---:|---:|
| A | none (baseline) | 0.475 | 9/20 |
| E1_section_only | control chunking, section path only | 0.525 | 10/20 |
| E2_document_section | control chunking, document title + section path | 0.475 | 9/20 |
| B | provider + document + section | 0.475 | 9/20 |

The ordering matches the mechanism exactly: the more constant the field, the more df inflation and the worse the result. Only the section path, which actually varies chunk to chunk, carries information — and it improves the baseline by **one case**, on a 20-case development set. That does not license adopting enrichment.

## Updated root-cause hypothesis

Two hypotheses have now been tested with controlled interventions and neither survives: **oversized chunks hide evidence** (falsified by EXP-005, zero rescued) and **missing structural lexical context** (falsified here, Δ0.000).

The surviving hypothesis is what AN-003 has pointed at all along: **BM25 cannot bridge the vocabulary gap between how a question is phrased and how the documentation states the answer.** Enrichment can only add terms that already exist in the document's structure. It cannot add the words a user actually used.

## Reproducibility validation

- EXP-006A reproduces EXP-000 (0.475, 9/20) and EXP-006C reproduces EXP-005A (0.500, 9/20), with identical per-case recall and identical hit ordering.
- `idx_chunk_search_vector` still used — 2 bitmap index scans. The EXP-005 planner regression has not returned.
- Repeated runs are byte-identical; score rounding before sorting is retained.
- All four configurations pass evidence-mapping validation, 22/22 spans, 0 offset mismatches.
- EXP-000 through EXP-005 artifacts untouched.

## Limitations

1. **n = 20 is development scale.** One case is 5 points of macro recall. Every result here is a 1–2 case movement; nothing is statistically significant.
2. **The field ablation was selected on the development set.**
3. **V3's table row-group mechanism was not tested** and remains the leading explanation for its AN-010 rescue.
4. **EXP-NULL still has not run** — retrieval lift over the model's own knowledge remains unknown.
5. **Dense retrieval has not been disproven.** Earlier dense numbers used an offline TF-IDF+SVD substitute, not a pretrained model.
6. **Section paths are mostly good** — 19 of 22 evidence spans sit under technical paths. The null result is not explained by useless headings.

## What is justified next

**Real pretrained dense retrieval.** Both lexical hypotheses have been tested and rejected. The remaining diagnosis — vocabulary mismatch — is exactly what a pretrained embedding model addresses, and AN-003 is its canonical test case. Blocked only on network egress.

Secondary and cheap: test V3's table row-group mechanism in isolation, and add stemming as a *third ranked list* rather than a replacement, so identifier precision survives.

**Not justified:** a reranker (AN-003 is absent from depth 300 in three of four configurations — a reranker reorders candidates, it cannot retrieve missing evidence), or freezing enrichment. **Baseline unchanged:** control chunking, no enrichment.

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